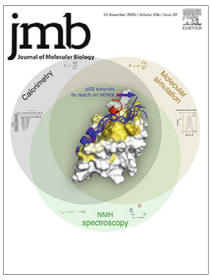
Research Article



Structural Analysis of Mammalian Sialic Acid Esterase

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Abstract

Sialic acid esterase (SIAE) catalyzes the removal of *O*-acetyl groups from sialic acids found on cell surface glycoproteins to regulate cellular processes such as B cell receptor signalling and apoptosis. Loss-of-function mutations in SIAE are associated with several common autoimmune diseases including Crohn's, ulcerative colitis, and arthritis. To gain a better understanding of the function and regulation of this protein, we determined crystal structures of SIAE from three mammalian homologs, including an acetate bound structure. The structures reveal that the catalytic domain adopts the fold of the SGNH hydrolase superfamily. The active site is composed of a catalytic dyad, as opposed to the previously reported catalytic triad. Attempts to determine a substrate-bound structure yielded only the hydrolyzed product acetate in the active site. Rigid docking of complete substrates followed by molecular dynamics simulations revealed that the active site does not form specific interactions with substrates, rather it appears to be broadly specific to accept sialoglycans with diverse modifications. Based on the acetate bound structure, a catalytic mechanism is proposed. Structural mapping of disease mutations reveals that most are located on the surface of the enzyme and would only cause minor disruptions to the protein fold, suggesting that these mutations likely affect binding to other factors. These results improve our understanding of SIAE biology and may aid in the development of therapies for autoimmune diseases and cancer.

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Introduction

Sialic acids are a diverse family of nine-carbon sugars which commonly cap the terminal ends of glycoproteins and glycolipids. There are over 80 naturally occurring sialic acid derivatives arising from *N*- and *O*-modifications (acetylation, methylation, sulfation, lactylation, and phosphorylation), and further diversity is introduced by different glycosidic linkages to the preceding sugar (α2-3/6/8).[1] This structural diversity determines their versatility as ligands for various receptors, thereby regulating a wide range of cellular processes.[2] Among mammals, the most abundant sialic acids are *N*-acetylneuraminic acid (Neu5Ac) and *N*-glycolylneuraminic acid (Neu5Gc), however humans cannot produce Neu5Gc due to a genetic defect.[3,4] The widespread modification of these sialic acids by *O*-acetylation is essential for development and regulation of the immune system.[5–7] *O*-acetylation itself is regulated by sialic acid *O*-acetyltransferases (SOATs) and sialic acid esterases (SIAEs; also known as sialate *O*-acetylesterases). These enzymes can differ between species resulting in different sialic acid derivatives. For example, in humans *O*-acetylation

can occur at positions C7 and C9, with C9 being more prevalent, whereas in other animals it's also observed at C4.[8]

In particular, 9-*O*-acetylation of Neu5Ac is prominent for its role in regulating B cell receptor (BCR) signalling.[9,10] This regulation is mediated by sialic-acid-binding immunoglobulin-like lectins (Siglecs), specifically CD22. CD22 recognition of Neu5Ac on BCRs leads to the inhibition of BCR signalling.[11] This recognition is blocked by 9-*O*-acetylation and unmasked by SIAE.[12] Inactivation of SIAE results in the accumulation of 9-*O*-acetylated Neu5Ac, uncontrolled BCR signaling and the development of self-reactive B cells.[13] Moreover, catalytically or secretion defective variants of SIAE have been associated with several common autoimmune diseases such as Crohn's, ulcerative colitis, and arthritis.[14–17] However, this association has been contested by other studies,[18–22] suggesting that further investigation is required.

O-acetylation is also a key regulator of apoptosis. The sialylated ganglioside GD3 induces apoptosis by accumulating in the mitochondrial membrane and triggering membrane permeabilization,[23] whereas 9-*O*-acetylation blocks translocation to the mitochondrial membrane.[24] In accordance with apoptosis resistance being a hallmark of cancer, 9-*O*-acetylated GD3 is abundant in several cancers and promotes their survival.[25–27] As such, induced SIAE expression promotes apoptosis in cancer cells and sensitizes them to apoptotic drugs, demonstrating SIAE's potential as a cancer therapy.[24,25,28,29] In some cases, *O*-acetylation may be unfavourable for cancer cells, as *O*-acetylation blocks immunosuppressive Siglecs. Blocking the immunosuppressive signalling of Siglec-7 and Siglec-15 increases anti-cancer immune responses and limits tumor growth in mouse models,[30,31] suggesting that inhibition of SIAE may be therapeutic in some cancers.

SIAE is secreted into the extracellular space where it can act upon extracellular sialoglycans,[32,33] but has also been suggested to localize in lysosomes to aid in the recycling of *O*-acetylated sialoglycans.[34] *O*-acetyl esterases from different species, such as horse and cow, have been shown to remove acetyl groups from 4- and 9-*O*-acetylated Neu5Ac[35,36] (Figure 1A). However, it is unclear how SIAE can accommodate such different substrates, as these modifications are on opposing ends of the Neu5Ac molecule, suggesting that substrates may bind in multiple configurations. Additionally, details regarding the catalytic mechanism of hydrolysis remain unknown. Although the enzyme is predicted to belong to the SGNH hydrolase superfamily, it lacks the conserved aspartic acid which completes the Ser-His-Asp catalytic triad. Instead, a predicted homology model suggested an unconventional catalytic triad of Ser-His-Thr.[32] To date, there is no reported experimental structural data of eukaryotic SIAE to confirm these results and provide insight on the enzyme's substrate specificity and catalytic mechanism.

Here, we report crystal structures of mammalian SIAE from three homologs. The structures reveal a central catalytic domain with a catalytic dyad instead of the expected catalytic triad. Crystal soaking experiments with SIAE's two substrates only yielded one of the hydrolyzed products, acetate, in the active site, but informed a proposed catalytic mechanism. Docking of triose sialoglycans with molecular dynamics suggests that the active site is broadly specific. Finally, structural mapping of disease mutations reveals that most do not disrupt the enzyme's fold or activity, but are clustered on the surface of the protein, suggesting a possible binding interface with regulatory factors. Together, these results provide structural insights into the function of mammalian SIAE.

Results

Overall structure of mammalian SIAE

SIAE is a 55 kDa protein that is expressed with an N-terminal signal peptide, targeting it to the endoplasmic reticulum where it is processed to be secreted or directed to lysosomes. In pursuit of an atomic resolution structure of mammalian SIAE, we recombinantly produced in insect cells, full-length constructs for dog (*Canis lupus familiaris*; 84% identity to human), guinea pig (*Cavia porcellus*; 81% identity), and rabbit SIAE (*Oryctolagus cuniculus*; 76% identity) and determined their X-ray structure to resolutions of 1.3 Å, 2.8 Å, and 3.0 Å, respectively (Figure 1B, C; Figure S1A; Table S1). All structures were solved by molecular replacement using AlphaFold2 predictions as search models.[37] Compared to our final models, AlphaFold accurately predicted the structures, with root-mean-square deviation (RMSD) values ranging from 0.590 to 0.697. Compared to one another, the three structures are highly similar, with RMSD values ranging from 0.390 – 0.559 Å. The most notable difference between the structures are two flexible loops surrounding the active site (Figure S1B). The flexibility of these loops may be required to accommodate different glycan chains downstream of the active site sialic acid (discussed below). The following structural analysis will mainly focus on dog SIAE since it has the highest sequence similarity to human and is the highest resolution structure.

The structures reveal a central catalytic domain, adopting an α/β/α sandwich, characteristic of SGNH hydrolases (Figure 1B, C). SGNH hydrolases are named after four blocks of conserved sequences which make up the catalytic residues: the nucleophilic serine (S), the glycine (G) and asparagine (N) of the oxyanion hole, and the histidine (H), accompanied by an aspartic

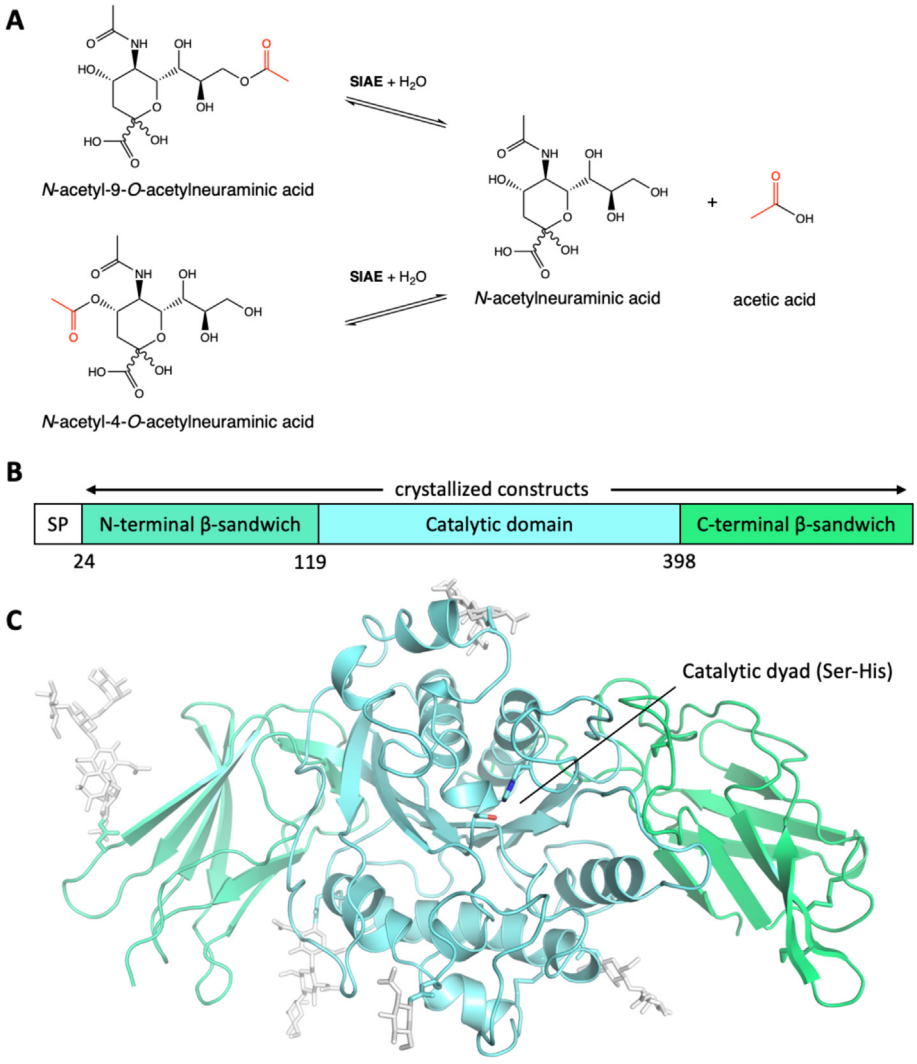


Figure 1. Crystal structure of mammalian SIAE. (**A**) SIAE *O*-acetylesterase activity. Depending on the species, SIAE catalyzes the removal of *O*-acetyl groups from *N*-acetyl-9-*O*-acetylneuraminic acid and *N*-acetyl-4-*O*-acetylneuraminic acid to yield *N*-acetylneuraminic acid and acetic acid. The acetyl group being removed is highlighted in red. The anomeric center (C2) is represented as an anomeric mixture. (**B**) Domain organization of SIAE. The enzyme is composed of a cleavable signal peptide (SP), a central catalytic domain, and N-terminal and C-terminal β-sandwich domains. For reference the diagram is numbered according to the human enzyme, however crystallized constructs were of the dog, guinea pig, and rabbit homologs. (**C**) Crystal structure of dog SIAE with the catalytic dyad (Ser128-His377), disulfide bonds, and *N*-linked glycans (white, for clarity) shown as sticks.

acid, responsible for activating the nucleophilic serine. The catalytic domain is flanked by N-terminal and C-terminal β-sandwich domains which are rigidly linked to the catalytic domain through extensive interfaces, burying ~2100 Å^2 and 2600 Å^2, respectively (Figure 1B, C). The function of the terminal domains remains unclear as their sequences do not share homology with any known domains. They do however structurally resemble the β-sandwich domains of the *Xanthomonas citri* xyloglucan acetylesterase,[38] a protein involved in the breakdown of the protective polysaccharide barrier of plant cell walls. It was proposed that these domains may serve as an extended platform for xyloglucan oligosaccharide anchoring during acetate removal,[38] although this idea has yet to be validated.

Several *N*-linked glycosylation sites are observed on the protein, predominantly on the catalytic domain but glycosylation sites vary on either β-sandwich domain depending on the species (Figure S1C). Only one predicted *N*-linked glycosylation site (guinea pig Asn402) was not observed due to poor electron density. The catalytic domain is stabilized by two disulfide bonds (Cys124–Cys197 and Cys236–Cys292)

and the C-terminal β-sandwich is stabilized by three disulfide bonds (Cys441–Cys469, Cys442–Cys447, and Cys485–Cys490).

Unexpectedly, the active site is revealed to be a catalytic dyad (Ser128-His377, Ser127-His377 in human) rather than the typical Ser-His-Asp/Glu catalytic triad found in esterases. Instead of an acidic residue completing the catalytic triad, the backbone carbonyl of Phe373 is responsible for stabilizing the orientation of the catalytic histidine His377 (Figure 2B), a mechanism seen in other SGNH hydrolases which also lack a catalytic aspartic acid[39–41] (discussed below). It was previously proposed that the enzyme utilizes an unconventional catalytic triad (Ser-His-Thr), based on their proximity to one another in a predicted homology model and abolished enzymatic activity upon mutation to alanine.[32] However, the corresponding

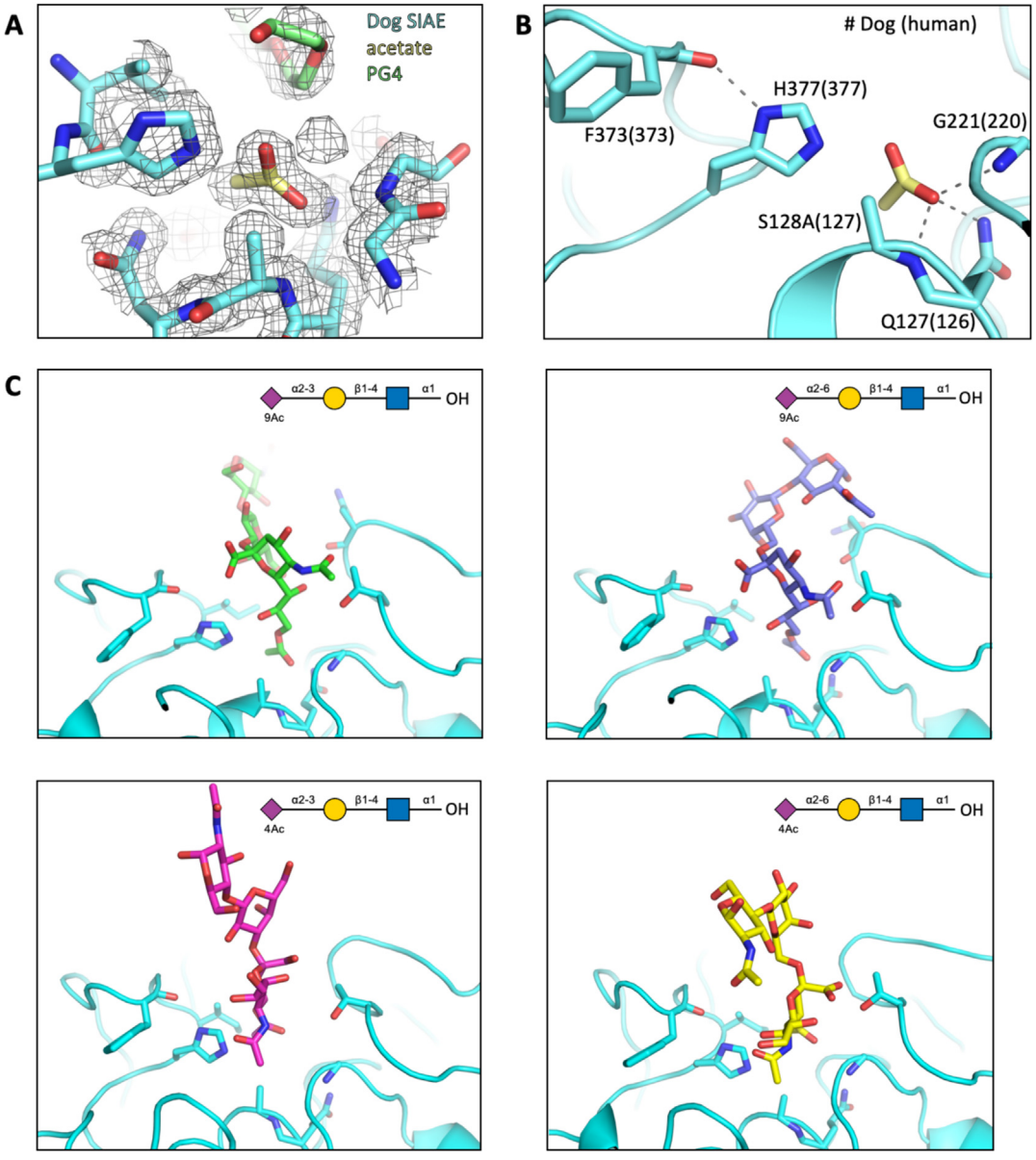


Figure 2. Active site of inactive S128A mutant dog SIAE bound to acetate. (**A**) The initial (pre-ligand placement) 2fo-fc electron density maps contoured at 1 σ for acetate, tetraethylene glycol (PG4), and active site residues. No density corresponding to the second hydrolysis product, *N*-acetylneuraminic acid, is observed. (**B**) Catalytic residues and their interactions with acetate and each other. Hydrogen bonds are shown as dashed lines and corresponding human residues are in parentheses. (**C**) Models of 9-*O* and 4-*O* acetylated sialoglycans with different glycosidic linkages (α2-3 or α2-6) docked to S128A cfSIAE by molecular dynamics simulations. The docked sialoglycans are represented by Symbol Nomenclature for Glycans (SNFG). Active site residues and residues on adjacent loops are shown as sticks.

threonine in our structure is 11 Å away from the active site histidine and tightly packed under a loop, making it an unlikely participant in catalysis (Figure S1D). Mutation of the threonine to alanine would remove a hydrogen bond with a backbone carbonyl of this loop and likely impacts enzymatic activity indirectly, by disrupting the packing of residues adjacent to the active site.

Crystal structure of acetate bound SIAE

To gain further insight into the catalytic mechanism and substrate specificity, we attempted to determine the structure of SIAE bound to its two structurally distinct substrates, *N*-acetyl-9-*O*-acetylneuraminic acid and *N*-acetyl-4-*O*-acetylneuraminic acid (Figure 1A). To achieve this, inactive mutants were generated by mutating the active site serine to alanine and crystals of SIAE homologs were soaked in solutions containing 25 mM of either substrate for varying time points (10 min, 1 h, 6 h). For both substrates and all three soaking times, only the hydrolyzed product acetate was observed in the active site, with no continuous electron density corresponding to the second hydrolysis product, *N*-acetylneuraminic acid (Figure 2A). It's unlikely that the mutant enzymes retained activity, since the mutant dog SIAE was confirmed to be inactive on the general *O*-acetyl esterase substrate, 4-nitrophenyl acetate (Figure S2A). However, background acid hydrolysis may have occurred despite storing substrates in buffered solutions. *N*-acetylneuraminic acid was detected in our *N*-acetyl-9-*O*-acetylneuraminic acid stock solution but not our *N*-acetyl-4-*O*-acetylneuraminic acid stock (Figure S2B). Nevertheless, a high concentration of *N*-acetylneuraminic acid was still in the soaking solution. The absence of electron density for *N*-acetylneuraminic acid could suggest the active site lacks a high degree of specificity to distinguish between variant *O*-acetylneuraminic acids, or the enzyme has a relatively low affinity for monosaccharide substrates.

Substrate docking

To probe the enzyme's substrate specificity for *N*-acetylneuraminic acid as well as downstream sugars in a glycan chain, we aimed to computationally model 9-*O* and 4-*O* acetylated sialoglycans in the active site. Among the most common triose chains to cap complex *N*-linked glycans are Neu5Ac-Gal-GlcNAc chains. As such, structures of *O*-acetylated Neu5Ac-Gal-GlcNAc chains with different glycosidic linkages (Neu5Acα2-3Gal and Neu5Acα2-6Gal), were generated using the Glycam server.[42] Multiple conformations for each glycan were generated based on possible rotamers and manually docked in Coot[43] using the bound acetate's coordinates as a constraint. For each glycan, the conformation which produced the least steric clashes was selected for molecular dynamics simulations in GROMACS[44] to reach a stable binding mode. Regardless of the *O*-acetylation site and glycosidic linkage, the enzyme does not form hydrogen bonds or salt-bridges with *N*-acetylneuraminic acid or the preceding sugars (Figure 2C). Given the lack of substrate-specific interactions and the high flexibility of the loops adjacent to the active site, SIAE may be broadly specific for *O*-acetylated sialoglycans. A broad substrate specificity would coincide with the extensive diversity of sialic acid derivatives and glycans in general.

Catalytic mechanism of substrate hydrolysis

Although we were unable to obtain a structure with intact substrates, the acetate bound structure provides insight into the catalytic mechanism of hydrolysis. The acetate molecule is positioned in proximity to the catalytic dyad, with one carbonyl oxygen oriented towards the oxyanion hole. The oxyanion hole is formed by the backbone amides of Gly221 and the catalytic Ser128, and the side chain amide of the adjacent Gln127 (Figure 2B), all of which are conserved among SIAE orthologues. In this case, Gln127 occupies the role of the conserved asparagine in most SGNH hydrolases. The reaction follows the canonical mechanism of Ser-His-Asp esterases[45] with the key difference being that the backbone carbonyl of Phe373 functionally replaces the aspartic acid in the charge relay system necessary for activation of the serine nucleophile (Figure 3). The backbone carbonyl of Phe373 serves as a neutral hydrogen bond acceptor to stabilize the orientation of His377, but unlike catalytic carboxylic acids, the backbone carbonyl cannot deprotonate His377 to form the appropriate tautomer to deprotonate Ser128. Instead, proton transfer to His377 results in a positively charged intermediate. This same mechanism has been reported in other SGNH hydrolases with Ser-His catalytic dyads.[39–41] This state is less stable than the neutral tautomer and likely impacts the catalytic rate. Deprotonation of Ser128 by His377 allows for nucleophilic attack on the acetyl carbonyl carbon resulting in the first tetrahedral intermediate. The resulting negative charge on the carbonyl oxygen of the tetrahedral intermediate is stabilized by the partial positive charges contributed by the oxyanion hole. Release of *N*-acetylneuraminic acid results in the formation of the acyl enzyme intermediate. Either the formation or release of the acyl enzyme intermediate may be the rate-limiting step.[39,45] Next, a water molecule binds the acyl enzyme intermediate, allowing for His377 to deprotonate the water. The resulting hydroxide attacks the carbonyl to form the second tetrahedral intermediate. Finally, release of acetic acid restores the free enzyme.

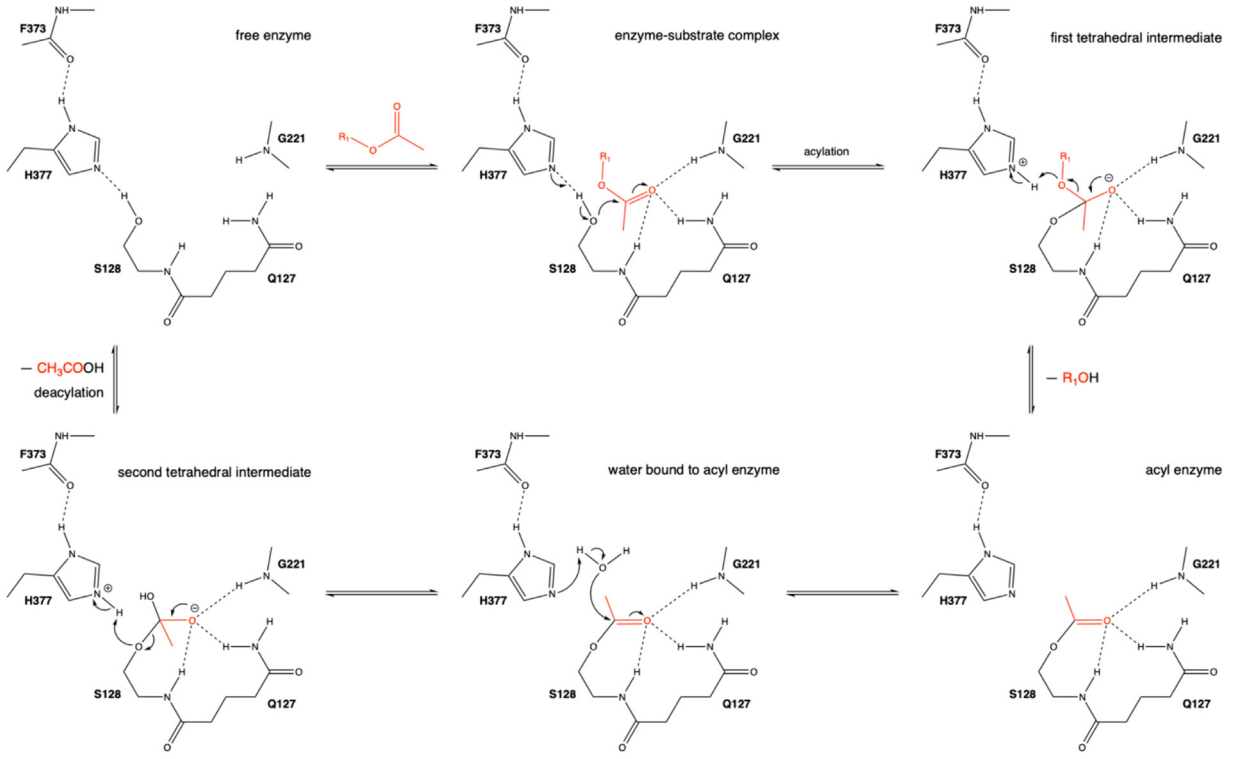


Figure 3. Catalytic mechanism of SIAE. SIAE undergoes a catalytic mechanism common to serine esterases. The substrate (where R_1 = *N*-acetylneuraminic acid linked at either position 4 or 9) and the resulting products are shown in red for clarity. Only the backbone carbonyl of Phe373 and the following amide are shown for clarity.

Disease mutations

Missense mutations in SIAE have been associated with several common autoimmune diseases such as juvenile idiopathic arthritis, rheumatoid arthritis, multiple sclerosis, Sjogren syndrome, systemic lupus erythematosus, type 1 diabetes, ulcerative colitis, and Crohn's disease.[14,15] These mutants were determined to either have defective catalytic activity, defective secretion, or both.[14,15] However, some of these mutations were found in control subjects[14,15] and the association in certain populations is under dispute.[18–20] Nevertheless, we mapped the known disease mutations onto our structure to understand the structural rationale behind defective catalytic activity and secretion. The analysis suggests that only a few mutations could cause major disruptions to the protein fold, whereas most mutations likely cause minor impacts on enzyme function as they are located on the surface of the protein and either do not form interactions with neighbouring residues or form minor interactions to stabilize the fold (Figure 4). The most severe fold disruptions include, C196F which disrupts a disulfide bridge responsible for stabilizing the catalytic core, possibly impacting catalytic activity, and G212R which results in a clash at the interface between the catalytic domain and the N-terminal β-sandwich domain. The implications of this interface disruption remain unclear as the function of the N-terminal β-sandwich domain is unknown. Although the remaining mutations appear to cause minor fold disruptions or have no obvious effect, it is interesting to note that they cluster on the face of the enzyme opposite to the active site. The observation that these mutations disrupt catalytic activity and secretion may suggest that this interface is involved in currently unknown interactions with regulatory factors, possibly those responsible for glycosidic processing and secretion of SIAE.

Discussion

The structures of mammalian SIAE determined here provide the first structural insight into the architecture of this distinct SGNH hydrolase and the hydrolysis of *O*-acetylated sialic acids, as well as provides hypotheses into the potential impact of mutations associated with autoimmune diseases. Still, questions remain regarding the function of the enzyme's auxiliary domains and substrate specificity.

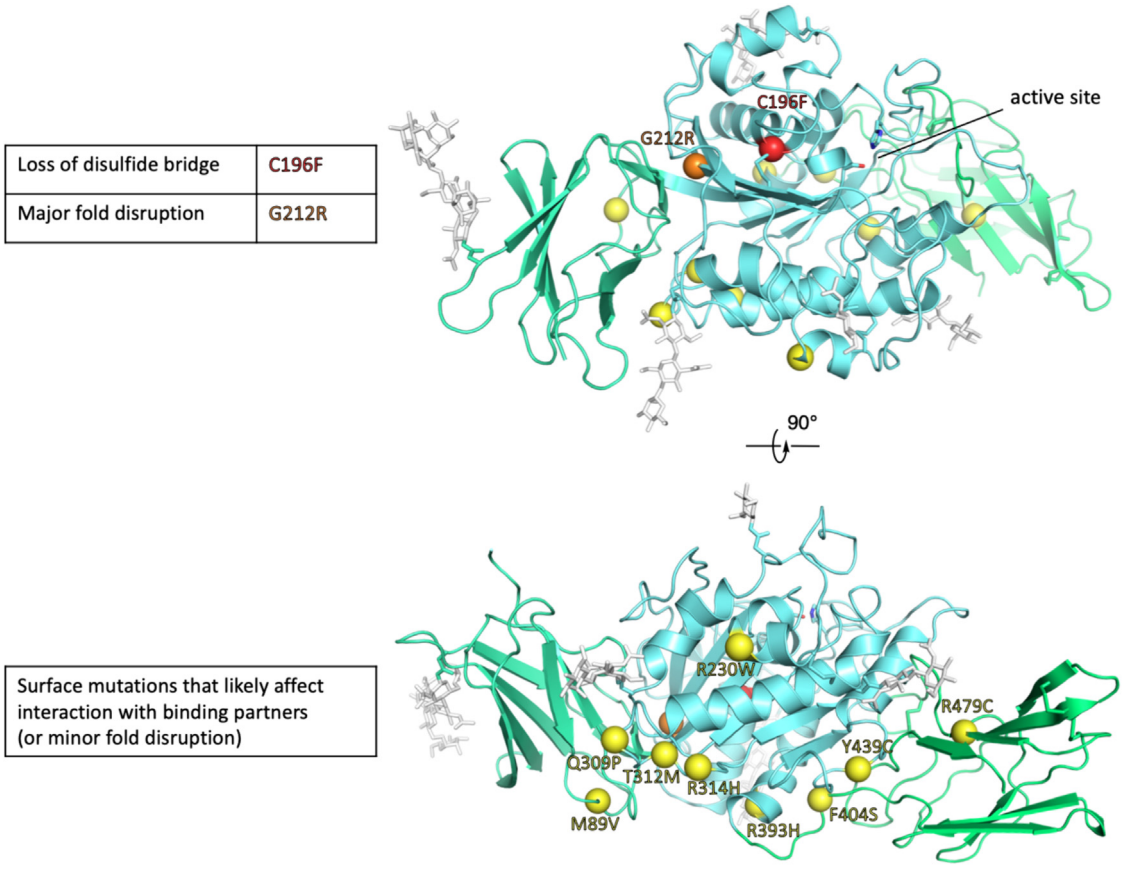


Figure 4. Structural mapping of mutations associated with autoimmune diseases. Locations of missense mutations in SIAE found in subjects with common autoimmune diseases, with disulfide bonds as sticks and *N*-linked glycans (white sticks) simplified for clarity. Mutations are coloured according to their structure-based predicted impact on the protein, from least to most severe (yellow to red).

Function of the β-sandwich domains

The structures reveal that the catalytic domain is flanked by N- and C-terminal β-sandwich domains, the function of which are unknown as they do not resemble any known domains at the sequence level. To further investigate their possible function, we searched the Protein Data Bank and the AlphaFold[37] database with the DALI[46] server to identify structurally similar domains. The top results were the N- and C-terminal β-sandwich domains of the *Xanthomonas citri* xyloglucan acetylesterase, an enzyme involved in the breakdown of plant cell wall xyloglucans which protect against pathogens.[38] The authors of the structure proposed that these domains may function as anchoring sites for xyloglucan oligosaccharides during deacetylation but did not provide supporting evidence for this hypothesis.[38]

Moreover, the majority of matches from our search were immunoglobulin-like (Ig-like) domains. Ig-like domains are β-sandwiches of seven or more strands organized into two sheets. Outside of immunoglobulins, they are found in numerous proteins with diverse functions and are often involved in interactions with other Ig-like domains but have also been associated with carbohydrate binding[47,48] and conferring stability in thermophilic organisms.[49–51] Our search only returned one esterase, from a thermophilic bacterium, whose Ig-like domain is involved in oligomerization, thermostability and activity.[51] In our study of SIAE, we found that the protein exists predominantly as a monomer, however the β-sandwich domains may still be involved in protein–protein interactions that have yet to be identified. For example, studies have suggested that SIAE may localize to the Golgi,[13,32,33] but without transmembrane anchors, SIAE may colocalize with a potential Golgi-resident protein mediated by the β-sandwich domains. Functionally, colocalization with other factors involved in the biosynthesis of *O*-acetylated glycans may serve a regulatory role. As for their impact on enzymatic activity, further investigation is required.

Substrate specificity

By attempting to solve the substrate bound structures, we aimed to determine how SIAE

confers specificity for differentially *O*-acetylated sialic acids. Unfortunately, we were only able to observe electron density for one of the hydrolysis products, acetate. The lack of electron density for the remaining product, *N*-acetylneuraminic acid, was surprising, and may suggest the enzyme has a low affinity for monosaccharide sialic acids or that it is not specific for *N*-acetylneuraminic acid. How certain species of SIAE are active on 9-*O* and 4-*O*-acetyl-sialoglycans remains unclear, as acetylation at the C4 position poses greater steric hinderance than C9. The C9 position is more accessible at the end of the glycerol-like side chain of the molecule, whereas the C4 position is in the center of the bulky neuraminic acid ring and adjacent to the 5-*N*-acetyl group. Manual docking of *O*-acetylated sialoglycans with different glycosidic linkages followed by molecular dynamics simulations, revealed that the enzyme lacks hydrogen bonding or ionic interactions specific for *N*-acetylneuraminic acid or downstream sugars. The flexibility of the active site may explain how SIAE can catalyze the removal of acetyl groups from both the C4 and C9 positions of Neu5Ac by allowing substrates to bind in different configurations. In addition, considering that sialic acids can have further *O*-modifications at other positions (C7, C8), SIAE may require a broadly accessible active site to be able to act upon diverse sialic acid derivatives. In fact, many carbohydrate acetyl esterases and SGNH hydrolases from bacteria and yeast have broad substrate specificities.[52–54] It's important to note that sialic acid derivatives differ across species[8] and *O*-acetylesterases of different species are specific for 9-*O* and 4-*O*-acetyl Neu5Ac or 9-*O*-acetyl Neu5Ac alone.[34–36,55] Insight gained from the SIAE homologs reported here may not directly translate to the human enzyme. Therefore, an extensive study of human SIAE's specificity for differentially modified sialic acids is needed.

Taken together, the results presented here provide the first structural characterization of mammalian SIAE. The acetate bound structure provides initial insight into the enzyme's catalytic mechanism and provides the foundation for future study of substrate specificity. The structures also shed light on the basis for catalytically and secretion defective variants associated with common autoimmune diseases to aid in the development of potential therapies.

Materials and Methods

Protein expression and purification

Recombinant SIAE was expressed as a secreted protein in *Sf*9 insect cells infected with baculovirus. The endogenous signal peptide was replaced by the melittin signal peptide MKFLVNVALVFMVVYISYIYA followed by a hexahistidine tag DRHHHHHHKL. Constructs encompassed residues 25–510 of dog SIAE (UniProt: A0A8C0LZX8), residues 24–518 of guinea pig SIAE (UniProt: H0VB40), and 24–514 of rabbit SIAE (UniProt: A0A5F9D0N0). The active site mutation S128A in dog SIAE was introduced to inactivate the enzyme for crystal soaking with substrates. All constructs and mutants were verified by sequencing. SIAE was isolated from expression culture medium by nickel-nitrilotriacetic acid (Ni-NTA) resin (Thermo Fisher Scientific) and further purified by size exclusion chromatography on a Superdex 200 increase column (GE Healthcare) in 10 mM tris-HCl (pH 8.1) with 100 mM NaCl. Eluted protein was concentrated to ~10 mg/ml, and flash-frozen.

Crystallization and data collection

Crystallization was attempted with ten mammalian homologs (*Bos taurus*, UniProt ID Q2KI90; *Canis lupus familiaris*, A0A8C0LZX8; *Cricetulus griseus*, G3IIB1; *Cavia porcellus*, H0VB40; *Homo sapiens*, Q9HAT2; *Mesocricetus auratus*, A0A1U7QMJ5; *Mus musculus*, P70665; *Oryctolagus cuniculus*, A0A5F9CAI5; *Rattus norvegicus*, P82450; *Sus scrofa*, A0A4X1VD39), but only four species yielded crystals which diffracted better than 4 Å (*Canis lupus familiaris*, *Cricetulus griseus*, *Cavia porcellus*, and *Oryctolagus cuniculus*). All crystals were grown by sitting drop vapor diffusion at 22 . Dog SIAE crystals were obtained in 16% (w/v) PEG 6000 with 10 mM tri-sodium citrate at 10 mg mL^{-1} and cryoprotected in well solution with 10% glycerol for 60 s, then in well solution with 20% glycerol for a few seconds. Guinea pig SIAE crystals were obtained in 10% (w/v) PEG 5000 MME and 12% (v/v) 1-propanol with 100 mM MES pH 6.5 at 10 mg mL^{-1} and cryoprotected in well solution with 10% glycerol for 60 s, then in well solution with 20% glycerol for 60 s, then in well solution with 30% glycerol for a few seconds. Rabbit SIAE crystals were obtained in 50% (v/v) MPD with 200 mM ammonium phosphate and 100 mM Tris pH 8.5 at 7.5 mg mL^{-1} and crystals were flash-frozen without cryoprotectant. Inactive mutant dog SIAE S128A crystals were obtained in 30% (v/v) PEG 400 with 100 mM sodium HEPES pH 7.5 at 10 mg mL^{-1}. Crystals were then incubated with *N*-acetyl-9-*O*-acetylneuraminic acid (Siachem SC-0006) or *N*-acetyl-4-*O*-acetylneuraminic acid (Biosynth MA29337) in soaking solution (75 mM sodium HEPES pH 7.5, 22.5% (v/v) PEG 300, 25 mM Tris base and 25 mM substrate) for 1 h at 22 and cryoprotected in 55 mM sodium HEPES pH 7.5, 16.5% (v/v) PEG 300, 25 mM Tris base, 25 mM substrate with 20% glycerol. X-ray diffraction data were collected at 100 K on beamline 08ID with an Eiger X 9 M detector at the Canadian Macromolecular Crystallography Facility (CMCF), Canadian Light Source.

Structure determination

Data were processed by HKL2000[56] with autocorrections enabled. This option applies several corrective procedures including ellipsoidal truncation which reduces high-resolution data completeness depending on anisotropy. This is made apparent in the number of reflections and completeness values in "Data collection" versus "Refinement" in Table S1. The wild-type SIAE structures were solved by molecular replacement by Phaser[57] in Phenix,[58] with search models derived from AlphaFold2 predictions.[37] Prior to molecular replacement, the isotropic B-factors of the search models were reset to 20. All structures were manually built in Coot[43] and refined by phenix.refine,[59] with anisotropic non-hydrogen B-factors applied for the wild-type and mutant dog SIAE structures and noncrystallographic symmetry (NCS) restraints applied for the guinea pig and rabbit SIAE structures. Crystallographic data collection and structure refinement statistics are presented in Table S1. Coordinates and structure factors have been deposited in the Protein Data Bank under the accession codes 8F9O, 8F9P, 8F9Q, 8F9R. Structural images were prepared with PyMOL (The PyMOL Molecular Graphics System, Version 2.5 Schrödinger, LLC) and glycan cartoons were prepared with Glycoglyph.[60] Sequence conservation figures were generated using ConSurf[61] with a multiple sequence alignment derived from a previously reported phylogenetic analysis.[32]

Enzymatic activity assay

Purified cfSIAE (7 nM) and the S128A mutant (5 μM) were incubated in triplicate with 1 mM 4-nitrophenyl acetate in buffer (10 mM Tris pH 7.2, 100 mM NaCl) for 1 h at 37 °C. The product, 4-nitrophenol, was quantified by absorbance at 405 nm and enzymatic activity was expressed as nmol product/second/nmol enzyme.

Mass spectrometry

Solutions of 1 mM *N*-acetyl-9-*O*-acetylneuraminic acid in 1 mM Tris Base and 100 μM *N*-acetyl-4-*O*-acetylneuraminic acid in 100 μM Tris Base were direct injected into a Bruker amaZon speed ETD ion trap mass spectrometer set to negative mode.

Molecular dynamics simulations

Structures of DNeup5Ac[9A]a2-3DGalpb1-4DGlcpNAca1-OH, DNeup5Ac[9A]a2-6DGalpb1-4DGlcpNAca1-OH, DNeup5Ac[4A]a2-3DGalpb1-4DGlcpNAca1-OH and DNeup5Ac[4A]a2-6DGalpb1-4DGlcpNAca1-OH were prepared using Glycam's carbohydrate builder tool[42]. Multiple conformations of each glycan were manually docked into the active site of dog SIAE S128A in Coot[43] based on the coordinates of the active site acetate. The conformations which posed the least steric clashes were selected for molecular dynamics simulation. The topology file of dog SIAE S128A was generated with the CHARMM36[62] force field and the topology files of the glycans were generated with the CHARMM General Force Field (CGenFF).[63] All system preparation (solvation and energy minimization) and simulations were performed in GROMACS.[44] The protein–ligand complexes were placed in the center of a dodecahedron box and solvated (TIP3P model) with an adjusted 10 Å distance from each side of the box. The system's charge was neutralized by the addition of five sodium ions. The energy minimization for each system (nsteps = 50 000, 100 ps) was done by the steepest descent algorithm. The Particle Mesh Ewald method was used to calculate electrostatic interactions.[64] The cut-off distances for the Van der Waals interactions, the Coulomb interactions, and the short-range neighbor list were all set to 12 Å.

Accession numbers

Coordinates and structure factors have been deposited in the Protein Data Bank under the accession codes 8F9O, 8F9P, 8F9Q, 8F9R.

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CRediT authorship contribution statement

Danilo Ide: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis. **Alexei Gorelik:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Katalin Illes:** Investigation. **Bhushan Nagar:** Writing – review & editing, Supervision, Funding acquisition.

Data Availability

Research data for this article is available on the Mendeley Data repository at data.mendeley.com/datasets/5nttxzp32w/1 and data.mendeley.com/datasets/s6jxg2pr4r/1.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal

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Appendix A. Supplementary material

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Abbreviations:
BCR, B-cell receptor; Gal, galactose; GlcNAc, *N*-acetylglucosamine; Ig-like, immunoglobulin-like; Neu5Ac, *N*-acetylneuraminic acid; Neu5Gc, *N*-glycolylneuraminic acid; RMSD, root-mean-square deviation; *Sf*9, *Spodoptera frugiperda* 9; SIAE, sialic acid esterase or sialate *O*-acetylesterase; Siglec, sialic-acid-binding immunoglobulin-like lectins

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